MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY
TTL, COUNTERS, CASCADABLE,
MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/320C, dated 6 March 1985, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

TITLE: Delete and substitute as printed above.

1.2.3, add the following new case outline:

"X C-2A (20-terminal, .350" x .350"), square chip carrier package".

1.3, Thermal resistance, junction-to-case (θ_{JC}), delete and substitute as follows:

"Thermal resistance, junction-to-case (θ_{JC}):
Cases A, B, C, D, 2, and X - - - - - - - - - - - - (MIL-M-38510, appendix C)".

1.3, Junction temperature (T_J) 3/, delete and substitute as follows:

"Junction temperature (T_J) 2".

Bottom of page, footnote 2/: Delete in its entirety. Footnote 3/: Delete "3" and substitute "2".

PAGE 5

TABLE II, Interim electrical parameters: Delete "pre burn-in".

TABLE II: Delete "Additional electrical subgroups for group C period inspections" in its entirety.

PAGE 6

4.2a.(1): Delete "D or E" and substitute "D, E, or F".

4.4.3b.(1): Delete "D or E" and substitute "D, E, or F".

4.4.3b.(3), delete and substitute the following:

"(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
13	Reprinted without change
14	14

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PAGE 8

FIGURE 1: Delete "Case 2" and substitute "Cases 2 and X" (two places).

PAGE 9

FIGURE 1: Delete "Case 2" and substitute "Cases 2 and X".

PAGES 24 THROUGH 29

TABLE III, terminal designations: Delete "Case 2 1/" and substitute "Cases 2, X 1/".

PAGE 30

Footnote 4/, delete and substitute the following:

"4/ A = 2.5 V, B = 0.4 V".

Footnote 6/, delete and substitute the following:

"6/ Output voltages shall be either:
a. = H $\geq$ 1.5 V
b. = L $\leq$ 1.5 V".

PAGES 32, 33, AND 34

TABLE III, terminal designations: Delete "Case 2 1/" and substitute "Cases 2, X 1/".

PAGE 35

Footnotes 8/ and 9/, delete and substitute as follows:

"8/ A = 2.5 V, B = 0.4 V

"9/ Output voltages shall be either:
a. = H $\geq$ 1.5 V
b. = L $\leq$ 1.5 V".

PAGES 36, 37, AND 38

TABLE III, terminal designations: Delete "Case 2 1/" and substitute "Cases 2, X 1/".

PAGE 38

Footnotes 8/ and 9/, delete and substitute as follows:

"8/ A = 2.5 V, B = 0.4 V

"9/ Output voltages shall be either:
a. = H $\geq$ 1.5 V
b. = L $\leq$ 1.5 V".

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1092)

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BDC count sequence
(See note A)

Count	Output			
	Q_D	Q_C	Q_B	Q_A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H

Biquinary (5-2)
(See note B)

Count	Output			
	Q_A	Q_D	Q_C	Q_B
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	H	L	L	L
6	H	L	L	H
7	H	L	H	L
8	H	L	H	H
9	H	H	L	L

Count sequence
(See note)

Count	Output			
	Q_D	Q_C	Q_B	Q_A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H
10	H	L	H	L
11	H	L	H	H
12	H	H	L	L
13	H	H	L	H
14	H	H	H	L
15	H	H	H	H

Reset/count function table

Reset inputs				Outputs			
$R_{0(1)}$	$R_{0(2)}$	$R_{9(1)}$	$R_{9(2)}$	Q_D	Q_C	Q_B	Q_A
H	H	L	X	L	L	L	L
H	H	X	L	L	L	L	L
X	X	H	H	H	L	L	H
X	L	X	L	Count			
L	X	L	X	Count			
L	X	X	L	Count			
X	L	L	X	Count			

Reset/count function table

Reset inputs		Outputs			
$R_{0(1)}$	$R_{0(2)}$	Q_D	Q_C	Q_B	Q_A
H	H	L	L	L	L
L	X	Count			
X	L	Count			

NOTE: Output Q_A is connected to input B.

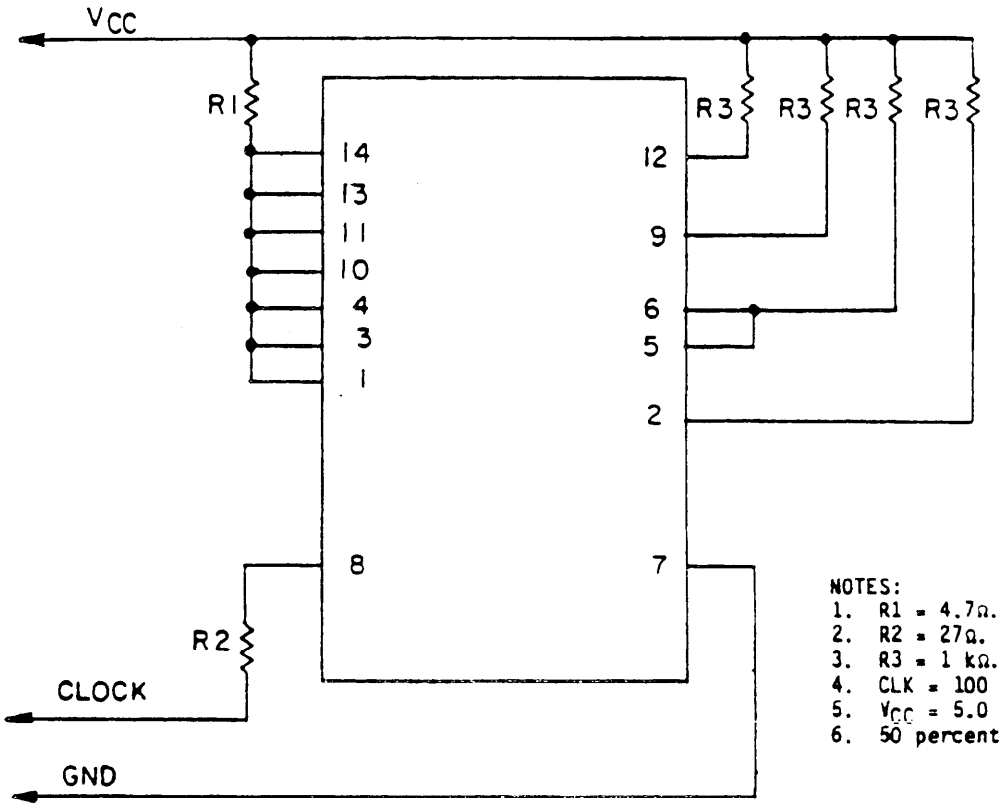
NOTES:
A. Output Q_A is connected to input B for BCD count.
B. Output Q_D is connected to input A for biquinary count.

FIGURE 3. Truth tables - Continued.

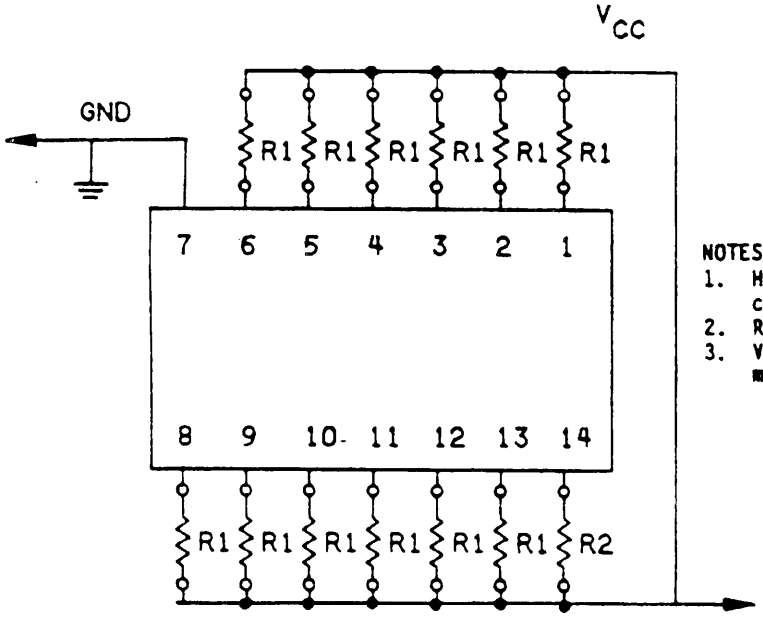
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Device types 01 and 02



NOTES:
1. R1 = 4.7Ω.
2. R2 = 27Ω.
3. R3 = 1 kΩ.
4. CLK = 100 kHz.
5. V_{CC} = 5.0 V minimum
6. 50 percent duty cycle.



NOTES (Device type 02, high termperature burn-in):
1. High temperature condition F (175°C) condition F burn-in circuit only.
2. R1 = 2 kΩ, R2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 4. Burn-in and life test circuit.

Supersedes page 14 of MIL-M-38510/320C
of 6 March 1985.

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